KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Mid Semester Examination, December, 2018
Third Year

TE 385: Signals and Systems *TIME: 1 hour*

Program of study: TE/BME INDEX NO:

Attempt all questions

SECTION A (10 Marks)

Fill in the blanks

1 For an odd signal $x(-t) =$ ..... for all t.

2 Discrete-time signal may be obtained by .... continuous-time signal

3 An LTI system is one which satisfies the properties of ..... and

.....

4 A shift of a signal to the right is a .....

5 The continuous- time unit impulse signal can be expressed as the

..... of the continuous- time unit step signal

6 The convolution of a signal with an impulse is equal to .....

7 If $x(t)$ is replaced by $x(at)$, then $x(at)$ is ..... version of $x(t)$ if $|a| > 1$

8 Discrete-time signals are ..... in time and

..... in amplitude

9 Any discrete-time signal can be expressed as ..... of time-shifted impulses

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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

Periodic

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Mid Semester Examination, December, 2018
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TE 385: Signals and Systems *TIME: 1 hour*

Program of study: TE/BME INDEX NO: 5934416

8/10

Attempt all questions

SECTION A (10 Marks)

Fill in the blanks

1 For an odd signal $x(-t)$ = ...$-x(t)$... for all t.

2 Discrete-time signal may be obtained by ...Sampling... ... continuous-time signal

3 An LTI system is one which satisfies the properties of ...Linearity... and Time-invariance

4 A shift of a signal to the right is a ...delay in time...

5 The continuous- time unit impulse signal can be expressed as the first derivative ... of the continuous- time unit step signal

6 The convolution of a signal with an impulse is equal to ...the original signal...

7 If $x(t)$ is replaced by $x(at)$, then $x(at)$ is ...linearly compressed... version of $x(t)$ if $|a| > 1$

8 Discrete-time signals are ...discrete... in time and continuous in amplitude

9 Any discrete-time signal can be expressed as ...Sum or sequence... of time-shifted impulses

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10 The Fourier series coefficients of discrete-time signals are ..... Periodic .....

SECTION B (50 Marks)

1 What is a periodic signal? Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x(t) = 4\cos(5t + \frac{\pi}{4})$

(b) $x[n] = 5\cos(5\pi n + \frac{\pi}{6})$

(c) $x[n] = e^{j5\pi n}$

2 Consider the system $y(n) = \sin x[n]$, state whether the system is

(a) Memoryless

(b) Linear

(c) Time invariant

(d) Causal

Justify your answers

3 Compute the convolution of $x(t) = u(t)$ and $h(t) = u(t)$

4 (a) If $y[n] = x[n] * h[n]$, show that $\sum_y = \sum_x \sum_h$ where $\sum_x = \sum_{n=-\infty}^{\infty} x[n]$ and $\sum_h = \sum_{n=-\infty}^{\infty} h[n]$.

(b) Compute $y[n] = x[n] * h[n]$ when $x[n] = \{\underset{\uparrow}{1}, 2, 2\}$ and $h[n] = \{\underset{\uparrow}{2}, 2, 1\}$ and check the correctness by using (a) above.

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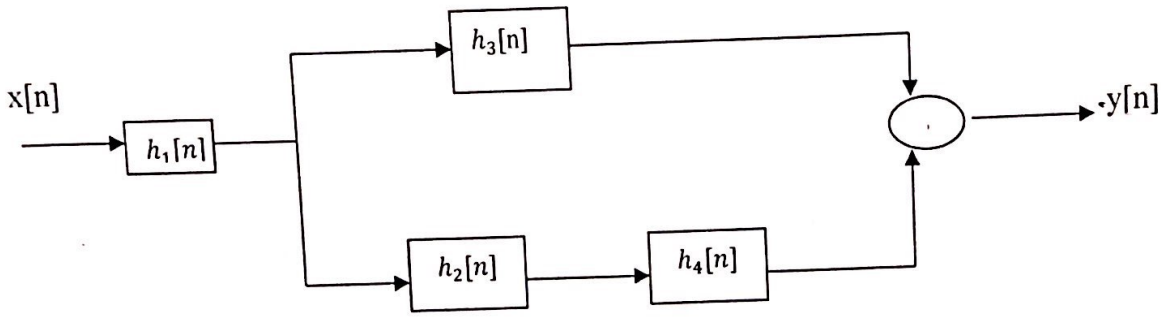
5 Consider the interconnection of LTI systems as shown below. Express the overall impulse response in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$ and $h_4[n]$.



6 Find the Fourier Series coefficient of a_k of the continuous time periodic signal

$$x(t) = \begin{cases} 2, & 0 \le t < 1 \\ 0, & 1 \le t < 2 \end{cases}$$

with fundamental frequency $\omega_o = \pi$.

7 Evaluate $\sum_{n=-\infty}^{\infty} e^{2n^2}\, \delta[n-2]$

8 Consider the signal $x[n] = \{0, 1, 1, \underset{\uparrow}{1}, 2, 2\}$

Sketch and label carefully the following: (a) $x[n-1]$ b) $x[3-n]$ c) $x[n]u[2-n]$

d) $x[n-1]\delta[n-3]$

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